

Congruence and Primes - Solutions

Problem 1. Let p be an odd prime. Let $a_1, a_2, \dots, a_p$ be a permutation of $1, 2, \dots, p$. Prove that there exist distinct integers $m, n \in \{1, 2, \dots, p\}$ such that $p \mid ma_m - na_n$.

Solution. If $a_k = p$ for some $k \neq p$, then we can choose $m = p$ and $n = k$ so that

$$pa_p - ka_k = pa_p - kp \equiv 0 \pmod{p}.$$

Now we assume $a_p = p$. Then $a_1, a_2, \dots, a_{p-1}$ is a permutation of $1, 2, \dots, p-1$. Therefore, we have $ka_k \not\equiv 0 \pmod{p}$ for each $k \neq p$. The assertion is the same as $ma_m \equiv na_n \pmod{p}$ for some $m \neq n$. If the assertion does not hold, then $\{ka_k : k = 1, 2, \dots, p-1\}$ covers all nonzero residue classes modulo p . In particular, we have

$$\prod_{k=1}^{p-1} (ka_k) \equiv \prod_{k=1}^{p-1} k \pmod{p}.$$

Since $a_1, a_2, \dots, a_{p-1}$ is a permutation of $1, 2, \dots, p-1$, this means

$$(p-1)!(p-1)! \equiv (p-1)! \pmod{p}.$$

By Wilson's theorem, this yields a contradiction $1 \equiv -1 \pmod{p}$. Therefore, we can always find $m \neq n$ such that $ma_m \equiv na_n \pmod{p}$.

Problem 2. Let p be a prime number such that $p \equiv 2 \pmod{3}$. Find

$$\prod_{k=1}^{p-1} (k^2 + k + 1) \pmod{p}.$$

Solution. The answer is 3.

The product is equal to $3 \prod_{k=2}^{p-1} \frac{k^3 - 1}{k - 1}$. Since $p \equiv 2 \pmod{3}$, $3 \nmid p-1$. Thus, $\{1^3, 2^3, \dots, (p-1)^3\}$ covers all nonzero residue classes modulo p (for example, let g be a primitive root modulo p . Then $\{g^3, g^6, \dots, g^{3(p-1)}\} \equiv \{g, g^2, \dots, g^{p-1}\} \equiv \{1, 2, \dots, p-1\} \pmod{p}$). It follows that

$$3 \prod_{k=2}^{p-1} \frac{(k^3 - 1)}{k - 1} \equiv 3 \prod_{k=2}^{p-1} \frac{k - 1}{k - 1} = 3 \pmod{p}.$$

Problem 3. (Belarus TST 2011 Test 3 Problem 1 modified) Let n be a fixed positive integer. Consider a natural number $a > 1$ and different odd prime numbers $p_1, p_2, \dots, p_n$ with

$$a^{p_1} \equiv 1 \pmod{p_2}, \quad a^{p_2} \equiv 1 \pmod{p_3}, \quad \dots, \quad a^{p_n} \equiv 1 \pmod{p_1}.$$

- (a) Prove that $p_i \mid (a - 1)$ for some $i = 1, 2, \dots, n$.
- (b) Can $(a - 1)$ be divisible by p_i for exactly one $i \in \{1, 2, \dots, n\}$?

Solution.

- (a) Suppose on the contrary that $a \not\equiv 1 \pmod{p_i}$ for each i . Let d be the order of a modulo p_2 . Since $a^{p_1} \equiv 1 \pmod{p_2}$, we have $d \mid p_1$. Also, $d \neq 1$ by assumption. As p_1 is a prime, we must have $d = p_1$. It follows that $p_1 \leq p_2 - 1$, and hence $p_1 < p_2$. Similarly, we get the contradiction $p_1 < p_2 < \dots < p_n < p_1$.
- (b) Yes. Consider an arbitrary prime p_1 . We choose any odd prime p_2 such that $p_2 \equiv 1 \pmod{p_1}$. Such a prime exists by Dirichlet's theorem. Since $p_1 \mid p_2 - 1$, the congruence $m^{p_1} \equiv 1 \pmod{p_2}$ has a solution $m \equiv r_2 \pmod{p_2}$ such that $r_2 \not\equiv 1 \pmod{p_2}$.

Similarly, after choosing some p_k , we choose any odd prime p_{k+1} such that $p_{k+1} \equiv 1 \pmod{p_k}$. The congruence $m^{p_k} \equiv 1 \pmod{p_{k+1}}$ has a solution $m \equiv r_{k+1} \pmod{p_{k+1}}$ such that $r_{k+1} \not\equiv 1 \pmod{p_{k+1}}$. We repeat the steps until we have chosen $p_1, p_2, \dots, p_n$.

Next, we can choose a such that $a \equiv r_k \pmod{p_k}$ for $k = 2, 3, \dots, n$, and $a \equiv 1 \pmod{p_1}$ by the Chinese remainder theorem. Then $a^{p_k} \equiv r_{k+1}^{p_k} \equiv 1 \pmod{p_{k+1}}$ for $k = 1, 2, \dots, n - 1$, and $a^{p_n} \equiv 1 \pmod{p_1}$. Also, $p_1 \mid a - 1$ but $p_k \nmid a - 1$ for $k > 1$.

Problem 4. Let a and b be distinct positive integers. Prove that there are infinitely many prime numbers p such that

$$p \mid a^n + b^{n+1}$$

for some positive integer n .

Solution. Let $d = (a, b)$, $a = ds$ and $b = dt$ so that $(s, t) = 1$. Then

$$a^n + b^{n+1} = d^n s^n + d^{n+1} t^{n+1} = d^n (s^n + (dt)t^n).$$

By letting $c = dt$, it suffices to show there are infinitely many primes p such that $p \mid s^n + ct^n$ for some n . Suppose on the contrary that only $p_1, p_2, \dots, p_k$ divide some of the terms. Note that $p_j \nmid t$ since otherwise $p_j \mid s, t$, contradicting $(s, t) = 1$. Also, if $p_j \mid s$, then $p_j \mid c$. WLOG assume $(p_1 p_2 \dots p_\ell, s) = 1$ and $p_j \mid s$ for $j = \ell + 1, \dots, k$.

Let M be a sufficiently large positive integer. Consider $n = \varphi(p_1^M p_2^M \dots p_\ell^M) + 1$. By the Euler-Fermat theorem, we have

$$s^n + ct^n \equiv s + ct \pmod{p_1^M p_2^M \dots p_\ell^M}.$$

Assuming $M > v_{p_j}(s + ct)$ for all $j = 1, 2, \dots, \ell$, this shows $v_{p_j}(s^n + ct^n) = v_{p_j}(s + ct)$. Also, for large M , we have $n \geq M$, so that

$$s^n + ct^n \equiv ct^n \pmod{p_j^M}$$

for $j = \ell + 1, \dots, k$. Thus, for such j , we find that $v_{p_j}(s^n + ct^n) = v_{p_j}(c)$. As the only prime divisors of $s^n + ct^n$ are $p_1, p_2, \dots, p_k$, we have

$$s^n + ct^n = p_1^{v_{p_1}(s+ct)} \cdots p_\ell^{v_{p_\ell}(s+ct)} p_{\ell+1}^{v_{p_{\ell+1}}(c)} \cdots p_k^{v_{p_k}(c)}.$$

This is a fixed number, but we can choose n large enough by considering large M . This is impossible unless $s = t = 1$, in which case we get the contradiction $a = b$.

Therefore, there are infinitely many prime numbers p dividing some of the terms.

Problem 5. (Japan MO 2012 Problem 3) Let p be a prime. Find all integers n such that for all integers x , if $x^n - 1$ is divisible by p , then $x^n - 1$ is divisible by p^2 .

Solution. The answer is all multiples of p .

Firstly, we show that $p \mid n$. Let g be a primitive root modulo p^2 , and consider $x = g^{p-1}$. Clearly,

$$x^n = g^{(p-1)n} \equiv 1 \pmod{p}.$$

Therefore, we must have $g^{(p-1)n} \equiv 1 \pmod{p^2}$. This implies $\varphi(p^2) \mid (p-1)n$, i.e. $p \mid n$.

Secondly, suppose $n = pm$ for some integer m , where p is an odd prime. If $x^n \equiv 1 \pmod{p}$, then $x^{pm} \equiv 1 \pmod{p}$, which yields $x^m \equiv 1 \pmod{p}$ by the Fermat little theorem. By the lifting the exponent lemma, we get

$$v_p(x^n - 1) = v_p(x^{pm} - 1) = v_p(x^m - 1) + v_p(p) \geq 1 + 1 = 2.$$

Therefore, $x^n \equiv 1 \pmod{p^2}$.

Lastly, suppose $p = 2$ and n is even. If $x^n - 1$ is even, then x must be odd, and it is well-known that $x^n - 1$ is divisible by 4.

Problem 6. (Iran MO 2015 3rd Round Number Theory Problem 5 modified) Let $p \geq 7$ be a prime number. Prove that one of the following numbers is in form of $x^2 + y^2$.

$$p + 1, 2p + 1, 3p + 1, \dots, (p-3)p + 1$$

Solution. Let $a \equiv 5^{-1} \pmod{p}$. There exist $x, y \in \left\{1, 2, \dots, \frac{p-1}{2}\right\}$ such that $x \equiv \pm 3a \pmod{p}$ and $y \equiv \pm 4a \pmod{p}$. Note that

$$x^2 + y^2 \equiv 9a^2 + 16a^2 = (5a)^2 \equiv 1 \pmod{p},$$

and $1 < x^2 + y^2 < 2\left(\frac{p}{2}\right)^2 = \frac{p^2}{2} < (p-3)p$. Therefore, $x^2 + y^2 = kp + 1$ for some $k = 1, 2, \dots, p-4$. In other words, $kp + 1$ is of the form $x^2 + y^2$ for some $k = 1, 2, \dots, p-4$.

Problem 7. (CMO 2011 Problem 6) Prove that for any given positive integers m, n , there exist infinitely many pairs of relatively prime positive integers a, b such that

$$a + b \mid am^a + bn^b.$$

Solution. We shall choose a and b such that $a+b = p$ is a prime. Then $(a, b) = (a, p-a) = (a, p) = 1$ since $0 < a < p$. We need $p \mid am^a + (p-a)n^{p-a}$, i.e. $p \mid am^a - an^{p-a}$. We shall choose p which is relatively prime to n . So this is the same as $p \mid (mn)^a - n^p$. By the Fermat little theorem, this means

$$p \mid (mn)^a - n.$$

The result is obvious when $m = n = 1$. When $mn > 1$, the sequence $mn, (mn)^2, (mn)^3, \dots$ consists of a finite number of prime divisors only, which are the prime divisors of mn . By Kobayashi's theorem, the sequence $mn - n, (mn)^2 - n, (mn)^3 - n, \dots$ consists of infinitely many prime divisors. In particular, there are infinitely many primes p such that $p \mid (mn)^k - n$ for some k but $p \nmid n$. Let $a \in \{1, 2, \dots, p-1\}$ such that $k \equiv a \pmod{p-1}$. By the Fermat little theorem, we have $p \mid (mn)^a - n$. Then we can define $b = p - a$ so that $a + b \mid am^a + bn^b$ from above.

Problem 8. (1st USEMO Problem 4) Prove that for any prime p , there exists a positive integer n such that

$$1^n + 2^{n-1} + 3^{n-2} + \dots + n^1 \equiv 2020 \pmod{p}.$$

Solution. We only consider $n = pm$ for some positive integer m . For each $k = 1, 2, \dots, n$, we write $k = pa + r$ where $0 \leq a \leq m-1$ and $1 \leq r \leq p$. Then we have

$$S = \sum_{k=1}^n k^{n+1-k} = \sum_{r=1}^p \sum_{a=0}^{m-1} (pa + r)^{n+1-(pa+r)} \equiv \sum_{r=1}^{p-1} \sum_{a=0}^{m-1} r^{m+1-a-r} \pmod{p}$$

by the Fermat little theorem. The inner sum is a geometric sequence. Thus,

$$S \equiv m + \sum_{r=2}^{p-1} r^{2-r} \cdot \frac{r^m - 1}{r - 1} \pmod{p}.$$

Now, we choose m such that $m \equiv 2020 \pmod{p}$ and $m \equiv 0 \pmod{p-1}$. Such an m exists by the Chinese remainder theorem. It follows that

$$S \equiv 2020 + \sum_{r=2}^{p-1} r^{2-r} \cdot \frac{1 - 1}{r - 1} \equiv 2020 \pmod{p}.$$